

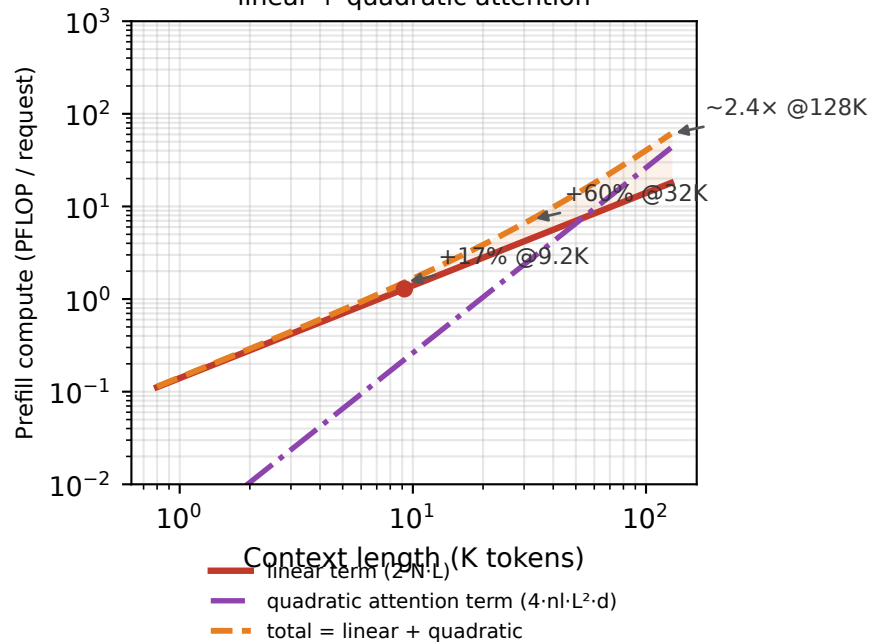


LEFT: PER REQUEST | RIGHT: PER GENERATED TOKEN

DO NOT COMPARE Y-AXIS MAGNITUDES DIRECTLY — different units, different baselines

Left = total prefill compute for ONE request (grows with context L). Right = compute for ONE generated token (fixed 2N, context-independent). The quadratic term is the textbook $4 \cdot n_{\text{layers}} \cdot L^2 \cdot d$ (matches Chapter 8). [ILLUSTRATIVE][DERIVED]

Prefill per request (decomposed):
linear + quadratic attention



Decode per token:
fixed 2N (context-independent)

